

Planar Kepler Conics, Collision Incompleteness, and the Levi-Civita Configuration Boundary

Route-A structural certificate HCS-C216

Abstract

We close a convention-locked theorem package for the attractive planar Kepler Hamiltonian. Runge–Lenz identities classify every nonzero-angular-momentum orbit as an ellipse, parabola, or hyperbola; the negative-energy period and radial action and the positive-energy scattering angle are normalized exactly. The zero-angular-momentum branch reaches the removed collision in finite physical time. Levi–Civita’s fixed-energy change of variables gives a smooth configuration equation, while a time-period resonance fixes a continuum rather than isolated primitive orbits. The radial action is written both as a closed-cycle integral and as its turning-point half-cycle, and all three collision antiderivatives are displayed.

Keywords: Kepler problem; Runge–Lenz vector; collision regularization; radial action; fixed-point continuum; Route A.

1 Frozen Hamiltonian and invariants

Identify the plane with $\mathbb{C}$ only when the Levi–Civita map is introduced. For $q \in \mathbb{R}^2 \setminus \{0\}$, $p \in \mathbb{R}^2$, and $\mu > 0$, set

$$H(q, p) = \frac{|p|^2}{2} - \frac{\mu}{r}, \quad r = |q|.$$

Hamilton’s equations are $\dot{q} = p$ and $\dot{p} = -\mu q/r^3$. Along a collision-free solution define

$$E = H, \quad L = q_1 p_2 - q_2 p_1, \quad A = (|p|^2 - \mu/r)q - (q \cdot p)p. \quad (1)$$

Differentiation gives $\dot{E} = \dot{L} = \dot{A} = 0$. Taking the scalar product and norm of A gives the two identities

$$A \cdot q = L^2 - \mu r, \quad |A|^2 = \mu^2 + 2EL^2. \quad (2)$$

2 All-energy conics, period, action, and scattering

Suppose $L \neq 0$, let $e = |A|/\mu$, and measure θ from A . Equation (2) is equivalent to

$$r(\theta) = \frac{L^2/\mu}{1 + e \cos \theta}, \quad e^2 - 1 = \frac{2EL^2}{\mu^2}. \quad (3)$$

Thus $E < 0$ is an ellipse (a circle when $e = 0$), $E = 0$ a parabola, and $E > 0$ a hyperbola. In the bound case put $a = -\mu/(2E)$ and $r_{\pm} = a(1 \pm e)$. Kepler’s period and the radial momentum are

$$P(E) = 2\pi\mu(-2E)^{-3/2}, \quad p_r^2 = 2\left(E + \frac{\mu}{r}\right) - \frac{L^2}{r^2}. \quad (4)$$

The normalization is part of the theorem, not a convention left implicit:

$$J_r = \frac{1}{2\pi} \oint p_r \, dr = \frac{1}{\pi} \int_{r_-}^{r_+} p_r \, dr = \frac{\mu}{\sqrt{-2E}} - |L|, \quad E < 0. \quad (5)$$

The turning-point substitution also gives $\partial_E J_r = P(E)/(2\pi)$. For $E > 0$, the asymptotic directions of (3) have $\cos \theta_{\infty} = -1/e$, so the unsigned hyperbolic deflection is

$$\chi = 2 \arcsin(1/e), \quad e > 1. \quad (6)$$

3 The radial collision boundary

When $L = 0$, the inward branch obeys

$$\dot{r}^2 = 2(E + \mu/r), \quad t_{\text{coll}}(r_0) = \int_0^{r_0} \frac{dr}{\sqrt{2(E + \mu/r)}} < \infty, \quad (7)$$

for every admissible starting radius. The endpoint $r = 0$ is absent from the physical phase space, hence the physical flow is incomplete. With $\alpha = -E > 0$, $u = \arcsin \sqrt{\alpha r_0/\mu}$ gives

$$t_{\text{coll}} = \frac{\mu}{\sqrt{2\alpha}^{3/2}} (u - \sin u \cos u);$$

for $E = 0$ it is $2r_0^{3/2}/(3\sqrt{2\mu})$, while for $E > 0$, with $u = \text{arsinh} \sqrt{Er_0/\mu}$, it is

$$t_{\text{coll}} = \frac{\mu}{\sqrt{2E}^{3/2}} (\sinh u \cosh u - u). \quad (8)$$

These formulas separate the collision boundary from the $L \neq 0$ conic classification; no collision is silently counted as a periodic orbit.

Revision focus. Round 1 adds the action normalization, all collision antiderivatives, and the physical incompleteness boundary.

Declarations

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